Score-constrained ensembles: action, records and the controller

A specified shared-score restriction of classical phase sheets gives the Fisher field action exactly. It survives canonical coordinate copies and retains their conditional covariance bound, but fails ordinary free transport and unrestricted terminal conjugate readout. A positive switching representation realizes the supplied field law on its explicit positive non-Gaussian forward orbit. Its force/rate controller determines the field pair within this architecture. Branch controllers retain the energy of unread preparations; a fresh marginal-score controller loses it.

These are conditional constructions, with $\kappa \geq 0$ supplied. Composition closure where proved selects no positive coefficient. Neither an autonomous energy-supplying apparatus nor the necessity of physical $\hbar > 0$ follows. The complete Gaussian recording and nonlinear-copy countertests are in the recording treatment. All historical model/coordinator checks were internal, with their corrections incorporated; no independent peer review or literature novelty is claimed.

Shared-score preparations and canonical records

The missing signed term in (117) can be generated by a positive classical preparation and the ordinary copy impulse. The preparation changes the initial correlations, including correlations invisible to the Gaussian covariance matrix. Full pointer readout and ordinary free transport give distinct failures. Thus it supplies a concrete realization of the conditional score allocation without promoting it to a physical law for all apparatus records.

Let $x = (q, z_1, \ldots, z_n)$ have a smooth strictly positive normalized configuration density $\rho(x)$, and let $S(x)$ be a real action field. Introduce one independent equally probable sign $\xi = \pm 1$, common to the body and all pointers, and set their canonical momenta to

$$p_a = \partial_a S + \kappa \xi \partial_a \log \rho, \qquad \kappa \geq 0. \tag{136}$$

This defines a positive classical probability measure on phase space, usually singular rather than a smooth full-dimensional density. Its additional premise is the particular configuration-dependent momentum preparation (136), with the same sign retained in every copy. Even when ρ factors, the full body–pointer phase-space law generally does not factor. At fixed configuration the mean momentum is ∇S; finite second moments give the actual body kinetic expectation $\int \rho S_q^2/(2m) + \kappa^2 I_q(\rho)/(2m)$.

Proposition 20 (copy covariance and its record scope). An impulsive Hamiltonian $G = f(q)\pi$ on one pointer, with its exposure absorbed into the smooth length-valued function f, acts as

$$\begin{aligned} q' &= q, & z' &= z + f(q), & p' &= p - f'(q)\pi, \quad \pi' = \pi, \\ \rho'(q, z') &= \rho(q, z' - f(q)), & S'(q, z') &= S(q, z' - f(q)). \end{aligned} \tag{137}$$

It preserves (136) exactly, including its sign. Finite compositions of these shears therefore preserve the class. Conditional on the entire final pointer-coordinate vector R, let $\rho_R(q)$ be the body density and $s_R(q) = \partial_q S'(q, R)$. Whenever the conditional second moments, Fisher information and integration-by-parts boundary terms exist,

$$\begin{aligned} E[p' \mid q, R] &= s_R(q), \\ \operatorname{Var}(p' \mid R) &= \operatorname{Var}_R(s_R) + \kappa^2 I(\rho_R), \\ \det \operatorname{Cov}((Q, p') \mid R) &\geq \kappa^2 \operatorname{Var}_R(Q) I(\rho_R) \geq \kappa^2. \end{aligned} \tag{138}$$

This is a coordinate-record statement. It does not include isolated pointer conjugates or assert free-motion closure.

Proof. The configuration shear in (137) has determinant one. The chain rule for either S or $\log \rho$ gives $\partial_{q'} = \partial_q - f'(q)\partial_z$ and $\partial_{z'} = \partial_z$, exactly the momentum transformation. The

same argument iterates. The sign remains independent of all final coordinates. Conditioning on R changes $\log \rho'$ by a normalization constant independent of q, so its body derivative is $\partial_q \log \rho_R$. Averaging the sign proves the first two lines of (138), and the body cross covariance is $\mathrm{Cov}_R(Q, s_R)$. Cauchy–Schwarz bounds its square by $\mathrm{Var}_R(Q)\,\mathrm{Var}_R(s_R)$. Finally $E_R[(Q - E_R Q)\partial_q \log \rho_R] = -1$ and another Cauchy–Schwarz inequality give $\mathrm{Var}_R(Q) I(\rho_R) \geq 1$. □

Actual weak-copy correction. Keep the weak-copy model's independent Gaussian *configuration* variables $Q \sim N(\mu, A)$ and $Z_i \sim N(0, 1/(rh_i))$, with $\sum_i h_i = T$. Replace its independent Gaussian momenta by (136) with constant S: $P = -\kappa\xi(Q - \mu)/A$ and $\Lambda_i = -\kappa\xi r h_i Z_i$. Their individual initial variances still equal κ^2/A and $\kappa^2 r h_i$. The frozen-body impulses give $R_i = Z_i + f(Q)$ and $Y_T = \sum_i h_i R_i$, so

$$P_T = -\kappa\xi \left[\frac{Q - \mu}{A} - r\{Y_T - T f(Q)\} f'(Q) \right] = -\kappa\xi \sigma_{Y_T}(Q). \tag{139}$$

The coordinate likelihood and its sufficient statistic remain (111). Consequently $\mathrm{Var}(P_T \mid R) = \kappa^2 I(\rho_{Y_T})$ and the cross covariance vanishes. This supplies (117)'s signed redistribution with the actual canonical recoil under every finite partition. It adds no independent posterior kick. The exact one-copy construction also applies to bounded nonlinear f.

For this frozen model the limiting coordinate path can be realized with $Y_t = t f(Q) + N_t$, N a Brownian motion of variance t/r, independent of (Q, ξ). Summing the same pointer momenta gives $\sum \Lambda_i = -\kappa\xi r N_T$ on every partition. The coordinate Brownian bridge is independent of (Q, ξ, N_T), so the entire declared coordinate history again gives (139)'s posterior. This observation uses the frozen coordinate and prescribed precision density; it establishes no adaptive or free-body continuation of this preparation.

Passive forgetting keeps the branch laws and their retained signs. Their actual kinetic energies sum, so (119) and (135), with $\hbar = 2\kappa$, remain the score and mean-flow variance excesses. Restarting from the marginal density's score would change the physical mixture. Conversely, relative constants of S are absent from (136)'s physical phase-space measure on disjoint supports. Their formal retention under the pullback is not a physical explanation of the reunion witness (122); a further physical memory would be needed to make them observable.

Two terminal conjugates reopen exact recovery. For two copies with $h_1, h_2 > 0$ and $\kappa > 0$, suppose all four classical pointer variables R_i, Λ_i can be read. Define $W_i = -\Lambda_i/(\kappa r h_i)$. Then $W_i = \xi Z_i$, and almost surely

$$\xi = \frac{R_1 - R_2}{W_1 - W_2}, \qquad f(Q) = R_i - \xi W_i. \tag{140}$$

The denominator is nonzero almost surely because $Z_1 - Z_2$ is a nondegenerate Gaussian. With the admissible affine copy $f(q) = q$, (140) recovers Q exactly and (139) recovers P_T; the complete-record conditional covariance is zero. In the final canonical variables these pointer variables Poisson-commute with the body pair. Ideal classical reading of both conjugates is therefore the extra access being tested, not a missing body recoil. The recovery is almost sure, not uniformly conditioned near equal pointer coordinates. Smooth measurements with vanishing preparation errors approach it on sets where the denominator is bounded away from zero. Restricting all terminal apparatus access to coordinate copies would be an additional physical premise.

Free transport already fails for one Gaussian body. Take $Q \sim N(0, A)$, $S = 0$, $P = -\kappa\xi Q/A$, mass $m > 0$, and ordinary Newtonian free motion. For $u = \kappa t/(mA)$,

$$\begin{aligned} Q_t &= (1 - \xi u) Q, \\ E Q_t^2 &= A(1 + u^2), \\ E Q_t^4 &= 3A^2(1 + 6u^2 + u^4). \end{aligned} \tag{141}$$

For $0 < |u| < 1$ the position law is the equal mixture of Gaussians with variances $A(1 - u)^2$ and $A(1 + u)^2$. Conditional on $Q_t = q$, the two retained signs have different weights for almost every

q: the two Gaussian branch densities differ. Thus the retained sign is no longer independent of configuration, as (136) requires. Passive sign retention does not repair closure. Away from $q = 0$, the conditional momenta $p_\xi(q) = -\kappa\xi q/[A(1 - \xi u)]$ are distinct. Their unequal weights also exclude a representation by any renamed independent fair sign, except on a set of position probability zero.

For comparison use the restricted Fisher field equations (4), derived below. Their coefficient is supplied as κ^2; ordinary free transport does not preserve this constraint. Hall–Reginatto's variational source and its reading level are given with that derivation.

Direct substitution gives a Gaussian solution of (4) from the same initial ρ, S:

$$\begin{aligned} B(t) &= A(1 + u^2), & \rho_t &= N(0, B(t)), \\ S(q,t) &= \frac{m\dot{B}(t)}{4B(t)}q^2 - \kappa \arctan u, & 2B\ddot{B} - \dot{B}^2 &= 4\kappa^2/m^2. \end{aligned} \tag{143}$$

Continuity gives $S_q = m\dot{B}q/(2B)$. Since $(\sqrt{\rho})_{qq}/\sqrt{\rho} = q^2/(4B^2) - 1/(2B)$, the quadratic and constant terms give the last identity and $\partial_t[-\kappa \arctan u] = -\kappa^2/(mB)$, respectively. The kinetic energy is $m\dot{B}^2/(8B) + \kappa^2/(2mB) = \kappa^2/(2mA)$, the same constant as in the Newtonian preparation. Nevertheless

$$E_{\text{Newton}}Q_t^4 - E_{\text{Fisher}}Q_t^4 = 3A^2[(1 + 6u^2 + u^4) - (1 + 2u^2 + u^4)] = \frac{12\kappa^2 t^2}{m^2}. \tag{144}$$

Both sides have length-fourth units. Agreement of variance and energy does not imply agreement of the transported density. The calculation uses only canonical variation and Gaussian moments, not an assumed quantum propagator.

1. What the classical sheet action supplies

Work on a connected position interval or the line. Require smooth positive densities, finite displayed energies, and vanishing boundary terms for the variations used below. A classical phase sheet has position density r and momentum W_q. Its canonical one-form and free Hamiltonian are $\int W\,\delta r\,dq$ and $\int rW_q^2/(2m)dq$, respectively, with $m > 0$.

This one-form follows from the ordinary particle form. If material coordinates a carry fixed probability mass, $q = X(a)$, $r\,dq = da$, and $v = \delta X \circ X^{-1}$ is a virtual displacement, then $\delta r = -\partial_q(rv)$. Integration by parts gives

$$\int W_q(X(a))\delta X(a)da = \int rW_q v\,dq = \int W\,\delta r\,dq. \tag{1}$$

Fixed total mass makes a spatially constant shift of W a gauge. This is a smooth variational identity; no infinite-dimensional well-posedness theorem is being assumed.

For two sheets of probability mass $1/2$, introduce the constraint

$$r_\sigma = \rho/2, \qquad W_\sigma = S + \kappa\sigma \log(\rho/\rho_{\text{ref}}), \qquad \sigma \in \{+1, -1\}. \tag{2}$$

Here ρ is normalized, S has action units, and the positive constant ρ_{ref} only makes the logarithm dimensionless. Changing it adds sheetwise constants and changes no momentum. Formula (2) is the field version of the supplied positive classical preparation in the shared-score preparation.

Proposition 1 (exact constrained action). Pulling the two-sheet one-form and energy back to (2) gives

$$\begin{aligned} \Theta &= \sum_\sigma \int W_\sigma \delta r_\sigma\,dq = \int S\delta\rho\,dq, \\ \mathcal{H} &= \sum_\sigma \int \frac{r_\sigma W_{\sigma,q}^2}{2m}dq = \int \frac{\rho S_q^2}{2m}dq + \frac{\kappa^2}{2m}I(\rho), \\ I(\rho) &= \int \frac{\rho_q^2}{\rho}dq. \end{aligned} \tag{3}$$

The first identity cancels the two score signs; the second expands the two squares. Thus variation of the *restricted* action $\int[\int S\rho_t\,dq - \mathcal{H}]dt$ yields

$$\rho_t = -\partial_q(\rho S_q/m), \qquad S_t = -\frac{S_q^2}{2m} + \frac{2\kappa^2}{m}\frac{(\sqrt{\rho})_{qq}}{\sqrt{\rho}}. \tag{4}$$

No further canonical-field bracket is needed after adopting the constraint and restricted variation. The Fisher modification is established prior art [@HallReginatto2002; reading: passages pp. 3–8, especially the classical action (4) and modified action/equations (17)–(18); source companion]. The calculation here identifies which correlated sheet restriction produces that action and tests its invariance.

Why this is not an invariant reduction. Unrestricted classical sheet continuity gives, on (2),

$$\partial_t(r_+ - r_-) = -\frac{\kappa}{m}\partial_q^2\rho. \tag{5}$$

For $\kappa > 0$, no strictly positive normalized density on the whole line has $\rho_{qq} \equiv 0$: an affine function cannot have those properties. Consequently every such constrained preparation departs from equal sheet densities immediately somewhere. Restricted variation and ordinary free sheet transport are different dynamics. Additional constraint reactions or state changes are required, as the exact Gaussian switching test in the Gaussian specialization also demonstrates. In particular (4)'s current uses the mean momentum S_q; it does not give both physical sheet velocities $W_{\sigma,q}/m$ without further dynamics.

On connected positive support the labeled sheet momenta determine S_q and hence S up to a common constant. At exact vacuum, separate constants on disconnected components become invisible to the ordinary physical phase-space measure. This construction cannot carry them into a vacuum limit merely by declaring them canonical data.

2. Population transport includes the reservoirs

Let h_L, h_R be nonnegative smooth normalized bumps with compact, ordered, disjoint supports. Between them fix a bridge $B = [q_L, q_R]$. Choose a smooth positive normalized density ρ_0 with finite Fisher information. Put

$$\begin{aligned} h &= h_R - h_L, & \rho_w &= \rho_0 + wh, \\ G(q) &= \int_{-\infty}^{q}(h_L - h_R)(y)dy, & M(w) &= \int \frac{G(q)^2}{\rho_w(q)}dq, \\ R_B &= \int_B \frac{dq}{\rho_0(q)}. \end{aligned} \tag{6}$$

Here w is probability mass transferred from left to right, $0 \le G \le 1$, $G = 1$ on B, and G vanishes outside the reservoirs and their connector. If K is their compact enclosing interval, $c = \min_K \rho_0 > 0$ and $H = \|h\|_\infty > 0$, then $|w| < c/(2H)$ ensures $\rho_w \ge c/2$ on K. Thus all terms in (6) are finite, with $M(w) \le 2|K|/c$, and smooth in this w range.

Vanishing exterior flux and continuity uniquely require $j(q) = \dot{w}\,G(q)$, because $\partial_q j = -\partial_t\rho_w$. The corresponding total transport kinetic energy is $mM(w)\dot{w}^2/2$. Both M and R_B have length-squared units. Smooth reservoir transport contributes positively, so $M(w) > R_B$.

Proposition 2 (continuity-compatible canonical family). Define an action field, up to a common constant, by

$$S_q(q; w, P) = \frac{PG(q)}{M(w)\rho_w(q)}. \tag{7}$$

The variable P has action units; it is not a point-particle momentum. Then

$$\begin{aligned} \Theta &= \int S\delta\rho_w \, dq = P \, dw, \\ H(w,P) &= \frac{P^2}{2mM(w)} + \frac{\kappa^2}{2m} I(\rho_w), \\ \dot{w} &= \frac{P}{mM(w)}, \qquad\qquad \dot{P} = \frac{P^2 M'(w)}{2mM(w)^2} - \frac{\kappa^2}{2m}\frac{dI(\rho_w)}{dw}. \end{aligned} \tag{8}$$

These local reduced Hamilton equations preserve H while w remains in the positive-density range and satisfy the ambient continuity equation exactly.

Proof. Since $h = -G_q$, integration by parts yields $\int Sh \, dq = \int S_q G \, dq = P$. The phase energy in (3) is $P^2/(2mM)$, proving the Hamiltonian and its equations. Their first equation gives $\rho_w S_q/m = \dot{w}G$, the required current. □

In this family the actual bridge action difference is

$$\Phi_B = S(q_R) - S(q_L) = \frac{R_B}{M(w)} P. \tag{9}$$

Thus Φ_B generally differs from the population-conjugate action $P = \int Sh_R dq - \int Sh_L dq$. The latter is a reservoir-weighted phase-action difference. Replacing it by bridge endpoint phases omits the reservoir motion. In coordinates (w, Φ_B) the canonical two-form has coefficient $M(w)/R_B$, rather than one.

For a fixed bridge density the exact dual identities are

$$E_{\text{transport},B} = \frac{mJ^2 R_B}{2}, \qquad \inf E_{\text{phase},B} = \frac{\Phi_B^2}{2mR_B}, \qquad J = \frac{\Phi_B}{mR_B}. \tag{10}$$

The phase infimum is the bridge lemma, (131), written with S instead of $\hbar\theta$; it is attained in the stated weighted H^1 class. The last identity makes the two energies equal. A low-density connector makes a fixed phase difference cheap and a fixed population flux expensive. It selects no absolute action scale.

The costs add over arbitrary spatial cuts, because both integrals of $1/\rho$ and of G^2/ρ add. This is an exact spatial transport identity, not Newton's temporal cell action. In the positive-tail family of recording (133), R_B grows as ϵ^{-2}. If reservoir costs additionally obey $M - R_B \le C$ uniformly, then $|P - \Phi_B| \le |P|C/R_B$ and the bridge becomes asymptotically dominant. That extra bound must be checked; exact vacuum is not such a check.

3. Continuity does not close the full field equation

Equations (8) are exact for the restricted model. They do not imply that its fixed density family is invariant under (4)'s Hamilton–Jacobi equation. For a Gaussian base $\rho_A = N(0, A)$ and initial $w = P = 0$, the full equation gives

$$\partial_t S_q = \frac{\kappa^2 q}{mA^2}. \tag{11}$$

The family (7) instead permits only a multiple of G/ρ_A at that initial time. It vanishes outside a compact interval, whereas (11) does not. This is an immediate nonzero transverse term for $\kappa > 0$. It cannot be removed by a spatially constant action gauge.

Adding Gaussian width supplies the missing global quadratic phase at $w = 0$. The next calculation tests whether that repairs closure for nonzero population transfer.

Proposition 3 (a fixed compact transfer shape cannot close). Fix $m, \kappa, A > 0$ and a nonzero $h \in C_c^\infty(\mathbb{R})$ with $\int h = 0$. The density family $\rho_{A,w} = \rho_A + wh$, allowing width variation and sufficiently small positive and negative w, cannot be locally invariant under (4) while including zero-phase-gradient states for every such w, satisfying continuity with vanishing

exterior flux, and having twice differentiable parameter trajectories. The same obstruction persists with Gaussian mean motion allowed. The claim concerns an open family of preparations, not all finite-dimensional models.

Proof. Set $r = \rho_A$ and $G = -\int_{-\infty}^{q} h$, so G is smooth and compactly supported. The integrated density tangents for width and transfer are $G_A = qr/(2A)$ and G. At an initially zero phase gradient, continuity first gives $\dot{A} = \dot{w} = 0$. Differentiating it and using (4) requires

$$\begin{aligned} \rho_w F_w &= c_A(w) G_A + c_w(w) G, \\ F_w &= \frac{2\kappa^2}{m} \partial_q \frac{(\sqrt{\rho_w})_{qq}}{\sqrt{\rho_w}}, \qquad c_A = m\ddot{A}, \quad c_w = m\ddot{w}. \end{aligned} \tag{12}$$

Outside the compact supports, $\rho_w = r$ and $F_w = \kappa^2 q/(mA^2)$; hence $c_A(w) = 2\kappa^2/(mA)$ for every small w. At $w = 0$ this already matches the full expression, giving $c_w(0) = 0$. Evaluation at any point with $G \neq 0$ makes $c_w(w)$ smooth, since the left side is smooth while $\rho_w > 0$.

Put $u = h/r$. Differentiating the square-root ratio gives $\delta[(\sqrt{\rho})_{qq}/\sqrt{\rho}] = (u'' - qu'/A)/2$. Retaining the variation of the density multiplier in (12), the necessary linearized identity is

$$u''' - \frac{q}{A} u'' - \frac{u'}{A} + \frac{q}{A^2} u = cg, \qquad g = G/r, \qquad c = m c_w'(0)/\kappa^2. \tag{13}$$

Since $G' = -h$, one has $u = qg/A - g'$. Substitution yields

$$\begin{aligned} -g'''' + \frac{2q}{A} g''' + \left(\frac{4}{A} - \frac{q^2}{A^2} \right) g'' - \frac{4q}{A^2} g' \\ + \left(\frac{q^2}{A^3} - \frac{1}{A^2} \right) g = cg. \end{aligned} \tag{14}$$

The function g is smooth and compactly supported because $r > 0$. All four initial data vanish at a point outside its support. Uniqueness for this regular fourth-order linear ODE forces $g \equiv 0$, hence $h = -G' \equiv 0$, a contradiction. Allowing the Gaussian mean adds a term proportional to r in (12); tail matching forces its coefficient to zero and leaves the same contradiction. □

The zero-phase-gradient and continuity assumptions are essential to the initial tangency test. The theorem permits isolated states, shapes depending on parameters or time, and other finite-dimensional families. It specifically rules out treating a fixed compact transfer bump plus Gaussian width as an exact free Fisher evolution. Generated shape information cannot be hidden inside that projection.

4. An exact positive nonlinear shape evolution

The compact-shape obstruction permits evolving tails. The following construction solves (4) exactly within its supplied dynamics. It also identifies a limitation of the first mode test: the linearized mode is a width/phase tangent, although its finite-amplitude completion is not Gaussian.

For $\kappa > 0$, introduce the auxiliary algebraic variable

$$\psi = \sqrt{\rho}\, e^{iS/(2\kappa)}, \qquad \psi_t = \frac{i\kappa}{m} \psi_{qq}. \tag{15}$$

The second equation is equivalent to (4) on positive density: its real part gives $\rho_t = -\partial_q(\rho S_q/m)$ and its imaginary part gives (4)'s second equation after differentiating the displayed product. It is a change of variables in the adopted field equations, not an independent physical propagation premise.

Fix $A > 0$ with length-squared units and put

$$\begin{aligned} u &= \frac{\kappa t}{mA}, & B &= A(1+u^2), & s &= \arctan u, \quad z = q/\sqrt{B}, \\ \psi &= B^{-1/4} e^{iuq^2/(4B)} \chi(z,s), & i\chi_s &= \left(-\partial_z^2 + \frac{z^2}{4}\right)\chi. \end{aligned} \tag{16}$$

For completeness, $a = \dot{B}/(2B) = \kappa u/(mB)$ and $b = \dot{s} = \kappa/(mB)$ obey $\dot{a} + a^2 = b^2$. Differentiating the second line cancels the terms $-a\chi/2$ and $-az\chi_z$ on both sides of (15). The remaining quadratic phase coefficient is $bz^2/4$, giving the last equation in (16). Thus no propagator formula is assumed.

Let

$$\begin{aligned} \phi_0(z) &= (2\pi)^{-1/4} e^{-z^2/4}, & H_2(z) &= (z^2-1)/\sqrt{2}, \quad \phi_2 = H_2\phi_0, \\ \psi_n(q,t) &= B^{-1/4} e^{iuq^2/(4B)} e^{-i(n+1/2)s} \phi_n(z), & n &\in \{0,2\}. \end{aligned} \tag{17}$$

Direct differentiation gives the oscillator eigenvalues $1/2$ and $5/2$ for ϕ_0 and ϕ_2. Gaussian moments give unit norms and orthogonality. Hence both displayed functions solve (15).

Proposition 4 (positive forward completion). For $0 < \epsilon < \sqrt{2}$, the normalized combination $\psi_\epsilon = (\psi_0 + i\epsilon\psi_2)/\sqrt{1+\epsilon^2}$ gives a smooth solution of (4) on the whole line for every $t \geq 0$:

$$\begin{aligned} \rho_\epsilon(q,t) &= \rho_G(q,t) \frac{1 + 2\epsilon \sin(2s) H_2(z) + \epsilon^2 H_2(z)^2}{1+\epsilon^2}, \qquad \rho_G = N(0,B), \\ S_\epsilon(q,t) &= \kappa\left(\frac{uq^2}{2B} - s\right) + 2\kappa \arctan \frac{\epsilon \cos(2s) H_2(z)}{1 + \epsilon \sin(2s) H_2(z)}, \\ \frac{\rho_\epsilon}{\rho_G} &\geq \frac{(1-\epsilon/\sqrt{2})^2}{1+\epsilon^2} > 0. \end{aligned} \tag{18}$$

The lower bound is relative to the Gaussian envelope; it supplies no absolute density floor at infinity. All displayed energies and moments are finite. The density is non-Gaussian for every $\epsilon > 0$.

Proof. Factoring out ψ_0 leaves $F = 1 + \epsilon[\sin(2s) + i\cos(2s)]H_2$. For forward time, $0 \leq s < \pi/2$, so $\sin(2s) \geq 0$; $H_2 \geq -1/\sqrt{2}$ gives $\Re F \geq 1 - \epsilon/\sqrt{2} > 0$. Thus the arctangent in (18) is one globally smooth phase, and $|F|^2$ gives its density and bound. Orthogonality gives exact normalization. Gaussian decay times a polynomial, with this strictly positive denominator, justifies the derivatives, moments and integrations by parts. The nonconstant quartic factor cannot be absorbed into a different Gaussian width. Equations (15)–(17) prove the field equations. □

The conserved energy and first two even moments are explicitly

$$\begin{aligned} E &= \frac{\kappa^2(1+5\epsilon^2)}{2mA(1+\epsilon^2)}, & EQ &= 0, \\ EQ^2 &= B\frac{1 + 2\sqrt{2}\epsilon\sin(2s) + 5\epsilon^2}{1+\epsilon^2}, \\ EQ^4 &= B^2\frac{3 + 12\sqrt{2}\epsilon\sin(2s) + 39\epsilon^2}{1+\epsilon^2}. \end{aligned} \tag{19}$$

Indeed (3)'s energy equals $2\kappa^2 \int |\psi_q|^2/m$. Integration by parts in (15) makes its time derivative zero. At time zero the imaginary coefficient cancels the derivative cross term, and $\int |\phi_0'|^2 = 1/4$, $\int |\phi_2'|^2 = 5/4$, giving E. For the moments use $E_0 z^2 = 1$, $E_0 z^2 H_2 = \sqrt{2}$, $E_0 z^2 H_2^2 = 5$, and $E_0 z^4 = 3$, $E_0 z^4 H_2 = 6\sqrt{2}$, $E_0 z^4 H_2^2 = 39$. Here E_0 integrates against ϕ_0^2. The variance has second derivative $4E/m$, as required by the free field virial identity; the linear-in-time term records the initial position/current correlation.

At first order in ϵ the paired tangents are

$$\delta\rho = 2\rho_G \sin(2s) H_2, \qquad \delta S = 2\kappa \cos(2s) H_2. \tag{20}$$

They are width and quadratic-phase tangents, not an independent linear population-transfer mode. The result is their exact non-Gaussian nonlinear completion. It carries evolving tails rather than a fixed compact transfer bump. Restricting this one solution to any time partition is exactly consistent; no general continuum existence theorem or separated-packet apparatus has been constructed.

Forward time matters: at $s = -\pi/4$ the factor is $1 - \epsilon H_2$ and has real zeros. Thus (18) does not define a globally positive real-field family under all backward evolution. For fixed t, $\kappa \downarrow 0$ gives a stationary non-Gaussian density, $S \to 0$, and $E \to 0$. The construction leaves the zero branch admissible.

5. Recording generates a shape outside the finite mode family

Free completion must survive an actual intervention. Couple a body state (ρ, S) from (18), or any positive state with the following finite integrals, to a pointer configuration $Z \sim N(0, C)$, $C > 0$, with zero pointer action. Prepare the joint momenta with the shared score sign (2), so the full phase-space law need not factor. Apply the canonical copy (137) in the canonical-copy construction, with smooth length-valued $f(q)$. In the algebraic variables its pullback is exactly

$$\Psi'(q, z) = \psi(q)\eta_C(z - f(q)), \qquad \eta_C(y) = (2\pi C)^{-1/4} e^{-y^2/(4C)}. \tag{21}$$

The complete pointer-coordinate record $R = r$ gives

$$\rho_r(q) = \frac{\rho(q) e^{-(r-f(q))^2/(2C)}}{\int \rho(y) e^{-(r-f(y))^2/(2C)} dy}, \qquad S_r(q) = S(q), \tag{22}$$

up to a record-dependent constant action. These are the exact conditional fields; the sign stays fair and independent of coordinates.

Proposition 5 (generated shape and retained recoil). For $f(q) = q^2/\ell$ with fixed length $\ell > 0$, every posterior (22) from (18) lies outside every finite Hermite–Gaussian expansion, even allowing a different Gaussian mean, width and quadratic phase. The mean body-energy increase is nevertheless exact:

$$\begin{aligned} \Delta E_{\text{body}} &= \frac{\kappa^2}{2mC} E_\rho[f'(Q)^2] = \frac{2\kappa^2}{mC\ell^2} E_\rho Q^2, \\ E_R I(\rho_R) - I(\rho) &= \frac{1}{C} E_\rho[f'(Q)^2]. \end{aligned} \tag{23}$$

All these posterior densities are smooth and positive, with finite Fisher information and body energy. The finite-mode failure is a failure of shape closure, not a violation of the conditional covariance bound (138).

Proof. The logarithmic posterior tail obeys $\lim_{|q|\to\infty} q^{-4} \log \rho_r(q) = -1/(2C\ell^2)$. A nonzero finite Hermite expansion over any Gaussian envelope has polynomial-times-Gaussian density and the same limit is zero. This proves the stated closure failure, without ruling out other finite models. For the energy differentiate (21): $\partial_q \Psi' = \psi_q \eta_C - f' \psi \eta_C'$. Integration in z kills the cross term since $\int \eta_C \eta_C' = 0$, and $\int |\eta_C'|^2 = 1/(4C)$. Multiplication by $2\kappa^2/m$ gives (23)'s first line, which is also the actual canonical recoil energy under (2). For its second line, the conditional score is $\partial_q \log \rho + (r - f) f'/C$; conditional on q, $r - f$ has mean zero and variance C. Expanding its square proves the identity. Quartic decay proves the stated posterior integrability. □

If the coordinate record becomes unread, the marginal position density and mean current return to the original ρ and $\rho S_q/m$. The actual body energy keeps the increment (23). Resetting

its momenta to that marginal density's two score sheets would erase precisely this energy. Here every conditional phase gradient is the same, so there is no additional current-variance excess; for general branches both excesses in recording (135) must be retained. The pointer energy is unchanged by this impulsive shear. The extra body energy therefore requires work from its actuator; a closed energy-supplying apparatus has not been derived. Full conjugate access remains the distinct terminal escape (140). None of these formulas selects $\kappa > 0$.

6. The two-sheet realization on any positive solution

The following balance equations hold on every smooth positive solution of (4); the explicit orbit (18) supplies a complete nonexplosion test. From here on $v = S_q/m$ is the mean velocity and

$$d = \frac{\kappa}{m}\partial_q \log \rho, \qquad p_\sigma = m(v + \sigma d), \qquad \sigma \in \{+1, -1\}, \tag{24}$$

so that $p_\sigma = W_{\sigma,q}$ in (2). A smooth external potential $U(q,t)$ is allowed: it adds $-U$ to the right side of (4)'s second equation and $\int \rho U\, dq$ to $\mathcal{H}$. The two sheet velocities $v \pm d$ are the forward and backward mean velocities of Nelson's stochastic mechanics with diffusion coefficient κ/m [@Nelson1966; reading: abstract, as in the stochastic route]. The realization below replaces his Brownian motion by one sign that switches at a finite rate, as the Gaussian switching specialization below does. The written derivations include the internal corrections to the original balance, controller and branch-protocol statements.

Proposition 6 (sheet field equations). On a smooth positive solution of (4) with potential U,

$$\begin{aligned} v_t + v v_q &= -\frac{U_q}{m} + \frac{\kappa}{m}\, d_{qq} + d\, d_q, \\ d_t + v\, d_q + d\, v_q &= -\frac{\kappa}{m}\, v_{qq}. \end{aligned} \tag{25}$$

Proof. Put $L = \log \rho$, so $(\sqrt{\rho})_{qq}/\sqrt{\rho} = \frac{1}{2}L_{qq} + \frac{1}{4}L_q^2$. Differentiating (4)'s second equation in q and dividing by m gives $v_t = -v v_q - U_q/m + (\kappa^2/m^2)(L_{qqq} + L_q L_{qq})$, which is the first line because $(\kappa/m)d_{qq} = (\kappa^2/m^2)L_{qqq}$ and $d d_q = (\kappa^2/m^2)L_q L_{qq}$. Continuity gives $L_t = -v_q - v L_q$; its q-derivative times κ/m is the second line. □

Proposition 7 (two-sheet realization: balance, force, energy). Let (Q_t, Σ_t) be a process on $\mathbb{R} \times \{\pm 1\}$ with velocity $\dot{Q} = v + \Sigma d$ between sign jumps, rate $\lambda_\sigma(q,t)$ from σ to $-\sigma$, and carried momentum $P_t = p_{\Sigma_t}(Q_t, t)$.

(i) *Balance.* With initial law $\rho(q,0)/2$ for each sign, a conservative forward evolution with unique paths preserves the law $\rho(q,t)/2$ at every time iff

$$\lambda_- - \lambda_+ = \frac{\kappa}{m}\frac{\rho_{qq}}{\rho}; \qquad \lambda_\sigma = \frac{\kappa}{m}\Big(-\sigma\frac{\rho_{qq}}{\rho}\Big)_+ \text{ is the minimal nonnegative choice.} \tag{26}$$

The balance identity alone establishes neither existence nor nonexplosion; Proposition 8 does so for the explicit orbit.

(ii) *Force and impulse.* Between jumps P changes by

$$F_\sigma = -U_q + \kappa d_{qq} + 2m\, d\, d_q - \sigma\kappa v_{qq} = -\partial_q V_\sigma, \qquad V_\sigma = U - \frac{\kappa^2}{m}\frac{\rho_{qq}}{\rho} + \sigma\frac{\kappa}{m}S_{qq}, \tag{27}$$

and at a jump $\sigma \to -\sigma$ by $\Delta p = -2\sigma m d = -2\sigma\kappa\, \partial_q \log \rho$. The symmetric part of the sheet potential is the rate difference in action units: $\frac{1}{2}(V_+ + V_-) - U = -\kappa(\lambda_- - \lambda_+)$.

(iii) *Energy.* With $K_\sigma = p_\sigma^2/(2m)$, pathwise $dK/dt = F_\Sigma \dot{Q}$ between jumps and $\Delta K = -2\Sigma\, mvd$ at a jump. In the mean, the κ-dependent part of the force and the jumps have powers

$$\begin{aligned} W_F^\kappa &= \int \rho\,[\kappa v d_{qq} + 2mvdd_q - \kappa d v_{qq}]\,dq, \\ W_J &= \sum_\sigma \int \frac{\rho}{2}\lambda_\sigma(K_{-\sigma} - K_\sigma)\,dq = \kappa \int v\,d\,\rho_{qq}\,dq, \qquad W_F^\kappa + W_J = 0, \end{aligned} \tag{28}$$

at every time, so that $\frac{d}{dt}E[K] = -\int \rho v U_q\,dq = d\mathcal{H}_{\rm kin}/dt$. Individual trajectories exchange energy with the actuator and at jumps; the ensemble mean does not.

Proof. (i) The forward equation for the density n_σ of $(Q, \Sigma = \sigma)$ is $\partial_t n_\sigma + \partial_q(n_\sigma(v + \sigma d)) = -\lambda_\sigma n_\sigma + \lambda_{-\sigma} n_{-\sigma}$. With $n_\sigma = \rho/2$ and continuity, the left side is $\frac{\sigma}{2}\partial_q(\rho d) = \frac{\sigma\kappa}{2m}\rho_{qq}$ and the right side is $\frac{\rho}{2}(\lambda_{-\sigma} - \lambda_\sigma)$. The two signs give one equation and its negative. Given a difference, the nonnegative solutions are the displayed pair plus a common nonnegative rate. (ii) Along the sheet flow, $dP/dt = m\,[\partial_t + (v + \sigma d)\partial_q]\,(v + \sigma d) = m[(v_t + v v_q) + \sigma(d_t + v d_q + d v_q) + d d_q]$; insert (25). For the potential, $\kappa d_q + m d^2 = (\kappa^2/m)(L_{qq} + L_q^2) = (\kappa^2/m)\rho_{qq}/\rho$ and $\kappa v_q = (\kappa/m)S_{qq}$. The jump changes σd to $-\sigma d$. (iii) The pathwise statements restate (ii). For the mean, apply the generator to $g = K_\sigma(q,t)$: $\partial_t g + (v + \sigma d)\partial_q g = (v + \sigma d)F_\sigma$, and $K_{-\sigma} - K_\sigma = -2\sigma m v d$, so $W_J = \int \rho\, mvd(\lambda_- - \lambda_+)\,dq = \kappa \int v d \rho_{qq}\,dq$. The identity $W_F^\kappa + W_J = 0$ is verified directly: with $\rho d = (\kappa/m)\rho_q$, $\rho_{qq} = \rho(L_{qq} + L_q^2)$ and $\rho_{qqq} = \rho(L_{qqq} + 3L_q L_{qq} + L_q^3)$,

$$W_F^\kappa + W_J = \frac{\kappa^2}{m}\int [v\rho_{qqq} - \rho_q v_{qq}]\,dq = 0$$

after two integrations by parts with vanishing boundary terms. Since $E[K] = \int \rho\,\frac{m}{2}(v^2 + d^2)\,dq = \mathcal{H}_{\rm kin}$, the last statement is the energy balance of (4) with potential. □

On the Gaussian (145) one has $v = ax$, $d = -bx$, $d_{qq} = v_{qq} = 0$ and $\rho_{qq}/\rho = (z^2 - 1)/B$, so (26)–(28) reduce to (146), (148) and (149): $F_\sigma = 2mb^2 x = 2\kappa^2 x/(mB^2)$, $\lambda_+ = b(1 - z^2)_+$ and $W_J = -2mab^2 B$. Three independent features are worth stating. The force is sign-dependent exactly through $-\sigma\kappa v_{qq}$, so it is sign-independent iff the mean velocity is affine; the Gaussian hid this. The force is a gradient of a potential that contains ρ_{qq}/ρ rather than the quantum potential $-(2\kappa^2/m)(\sqrt{\rho})_{qq}/\sqrt{\rho}$; the two differ by $-\frac{m}{2}d^2$, the sheet kinetic excess. And for a state with zero mean current ($v \equiv 0$, for instance a trapped ground state) both mean powers in (28) vanish and every jump has $\Delta K = 0$, although individual trajectories still exchange energy with the actuator: for $\rho = N(0, A)$ held by $U = \kappa^2 q^2/(2mA^2)$, with $S_t = -\kappa^2/(mA)$, one has $F_\sigma = \kappa^2 q/(mA^2)$ and velocity $-\sigma\kappa q/(mA)$, so $F_\sigma \dot{Q} = -\sigma\kappa^3 q^2/(m^2 A^3) \neq 0$ for $q \neq 0$. With a time-dependent U the total energy obeys $d\mathcal{H}/dt = \int \rho U_t\,dq$.

Proposition 8 (nonexplosion on the explicit orbit, all forward time). For (18) write $z = q/\sqrt{B}$, $s = \arctan u$, and $P(z,s) = 1 + 2\epsilon\sin(2s)H_2 + \epsilon^2 H_2^2 = |F|^2$, the factor of Proposition 4, so $\rho = \rho_G P/(1 + \epsilon^2)$. Then

$$\partial_z \log P = \frac{2\sqrt{2}\,\epsilon z\,(\sin 2s + \epsilon H_2)}{P}, \qquad \partial_z \operatorname{Arg} F = \frac{\sqrt{2}\,\epsilon z \cos 2s}{P}, \tag{29}$$

and in the clock s the sheet motion and rates read

$$\frac{dz}{ds} = -\sigma z + r_\sigma(z,s), \quad r_\sigma = 2\partial_z \operatorname{Arg} F + \sigma\partial_z \log P, \quad \frac{\lambda_\sigma}{b} = \Big(-\sigma B\frac{\rho_{qq}}{\rho}\Big)_+ \le c_\epsilon(1 + z^2), \tag{30}$$

with $|r_\sigma| \le r_\epsilon$ and c_ϵ finite constants depending only on ϵ; explicitly $|\partial_z \log P| \le \max\{4\sqrt{2}\epsilon/(1 - \epsilon/\sqrt{2}), 8/3\}$. Hence $|z(s)| \le (|z_0| + \frac{\pi}{2}r_\epsilon)e^{\pi/2}$ for all $t \ge 0$, the hazard integrated over the entire forward half-line is at most $\frac{\pi}{2}c_\epsilon(1 + \sup z^2) < \infty$ given Z_0, the sign jumps are almost surely

finitely many on $[0,\infty)$, and the process of Proposition 7 exists for all forward time with law (18). The orbit is even in q, so $v(0,t) = d(0,t) = 0$: the origin is invariant for both sheets and no trajectory crosses it.

Proof. With $H_2' = \sqrt{2}z$, $P_z = 2\epsilon H_2'(\sin 2s + \epsilon H_2)$ gives the first formula of (29). For the phase, $\partial_z \operatorname{Arg} F = (\operatorname{Re} F\, \partial_z \operatorname{Im} F - \operatorname{Im} F\, \partial_z \operatorname{Re} F)/P$ with $\operatorname{Re} F = 1 + \epsilon \sin(2s) H_2$, $\operatorname{Im} F = \epsilon \cos(2s) H_2$; the products of $H_2 H_2'$ cancel and leave $\epsilon \cos(2s) H_2'/P$. For the bound, note $P = (\sin 2s + \epsilon H_2)^2 + \cos^2 2s$, so $|\sin 2s + \epsilon H_2| \le \sqrt{P}$ and $|\partial_z \log P| \le 2\sqrt{2}\epsilon|z|/\sqrt{P}$. For $|z| \le 2$ use $P \ge (1 - \epsilon/\sqrt{2})^2$ from (18). For $|z| \ge 2$, $H_2 \ge 0$ and $\sin 2s \ge 0$ give $P \ge \epsilon^2 H_2^2$ with $H_2 \ge 3z^2/(4\sqrt{2})$, so $|\partial_z \log P| \le 16/(3|z|) \le 8/3$. The same two regions bound $\partial_z \operatorname{Arg} F$ and $\partial_z^2 \log P = P_{zz}/P - (P_z/P)^2$, where $P_{zz} = 2\sqrt{2}\epsilon(\sin 2s + \epsilon H_2 + \sqrt{2}\epsilon z^2)$. Now $S_q = \kappa u q/B + 2\kappa B^{-1/2} \partial_z \operatorname{Arg} F$ and $\partial_q \log \rho = B^{-1/2}(-z + \partial_z \log P)$, so with $a = ub$, $v = \sqrt{B}\,(az + 2b\,\partial_z \operatorname{Arg} F)$ and $d = \sqrt{B}\, b\,(-z + \partial_z \log P)$. Then $\dot z = (v + \sigma d)/\sqrt{B} - az = b[-\sigma z + 2\partial_z \operatorname{Arg} F + \sigma \partial_z \log P]$, and $ds = b\,dt$ gives (30). Also $B\rho_{qq}/\rho = (-1 + \partial_z^2 \log P) + (-z + \partial_z \log P)^2$, which is bounded by $c_\epsilon(1 + z^2)$. Grönwall on $(|z|)' \le |z| + r_\epsilon$ gives the growth bound; the clock runs only to $s = \pi/2$ over the whole physical half-line. Conditional on Z_0, successive hazard clocks are dominated by one Poisson clock of finite rate, as in the proof of Proposition 21 of the recording note, which proves existence and nonexplosion; uniqueness of the forward solution with locally bounded rates identifies the law as $\rho/2$ per sign. Evenness of (18) in q gives the invariance of the origin. □

Two limits of this realization are explicit. It needs $\rho > 0$: at a node, d and the rates diverge, and (18) is positive only forward in time. And it supplies no crossing, exactly as (145)–(148) did: packet crossing or relative-phase reunion cannot be tested on an even orbit. The posterior states (22) with the quadratic copy have $d \sim -(2\kappa/mC\ell^2)q^3$ at $t = 0$, so the $-$ sheet moves outward with cubic speed and would escape in finite time; its rate $\lambda_- \sim (m/\kappa)(2\kappa/mC\ell^2)^2 q^6$ has infinite integrated hazard before escape, so a switch to the inward $+$ sheet occurs almost surely first. Nonexplosion over an interval for these tails needs the evolved fields and is not claimed here.

The Gaussian switching specialization

The free failure (144) can be repaired mathematically without a negative transition rate. It requires both sign switches and a continuous force, and that force cannot be a preparation-independent law on the presently retained microscopic state. The following exact construction quantifies these requirements. It treats (143)'s Gaussian evolution as a target; it does not derive the Fisher Hamiltonian as physical free mechanics.

Fix $m, A, \kappa > 0$ and a constant mean μ. Write $x = q - \mu$ and

$$\begin{aligned} u_t &= \kappa t/(mA), & B(t) &= A(1 + u_t^2), \\ a(t) &= \dot B/(2B), & b(t) &= \kappa/(mB), \\ p_\sigma(q,t) &= m(a - \sigma b)x, & \sigma &\in \{+1, -1\}. \end{aligned} \tag{145}$$

This is (136) with $\rho_t = N(\mu, B(t))$ and $S_q = ma(t)x$. Require ordinary position velocity $\dot q = p_\sigma/m$ between locally finite sign jumps, with continuous position at a jump. The target joint configuration/sign density is $\rho_t(q)/2$ for each sign. If $\lambda_\sigma(q,t)$ is the rate from σ to $-\sigma$, its two forward equations require

$$\begin{aligned} \lambda_- - \lambda_+ &= b(z^2 - 1), & z &= x/\sqrt{B}, \\ \lambda_+ &= b(1 - z^2)_+, & \lambda_- &= b(z^2 - 1)_+. \end{aligned} \tag{146}$$

The second line is the minimal nonnegative choice; all other nonnegative choices satisfying the first add a common nonnegative rate to both. In particular a constant symmetric flip rate cannot repair the Gaussian score law.

Proposition 21 (exact positive path law). Put

$$s(t) = \int_0^t b(r)dr = \arctan u_t, \qquad \frac{dz}{ds} = -\sigma z, \qquad \ell_+ = (1-z^2)_+, \quad \ell_- = (z^2-1)_+. \tag{147}$$

Start with $Z_0 \sim N(0,1)$ and an independent fair sign. Between switches follow the displayed linear flow. Draw each next switch by its integrated hazard ℓ_σ along that flow, retaining the state at every observation. The process is nonexplosive, and the joint stationary density is $\varphi(z)/2$ for each sign, where φ is the standard normal density. Consequently $Q_t = \mu + \sqrt{B(t)}Z_{s(t)}$ has exactly the target density and independent fair sign for every $t \geq 0$. Sampling this one path at any time partition gives exact restriction consistency; refreshing signs at the cuts would be a different process.

Proof. The clock and coordinate change in (147) transform (145)'s velocity and (146)'s rates exactly. Along every path, $|Z_s| \leq |Z_0|e^s$; its rate is bounded by $1 + |Z_0|^2 e^{2s}$. This bound is finite through $s = \pi/2$. Successive hazard clocks can therefore be dominated by a finite-rate Poisson clock conditional on Z_0, proving existence and nonexplosion, even over the entire physical half-line. The stationary forward equation for the + sign reads $\partial_z(-z\varphi/2) = (z^2-1)\varphi/2 = (\ell_- - \ell_+)\varphi/2$; the other equation is its negative. Uniqueness of the locally bounded-rate path construction, or its finite-jump forward expansion, identifies this stationary law. The inverse coordinate change proves the Gaussian marginal. □

The switching mechanism has established sampling prior art. On either fixed sign of z, put $r = \log|z|$ and velocity $v = -\sigma$. Then $\dot{r} = v$ in clock s, the position density is proportional to $e^{r - e^{2r}/2}$, and the rate is $(v\Psi'(r))_+$ with $\Psi(r) = e^{2r}/2 - r$. This is the one-dimensional canonical Zig-Zag process [@BierkensFearnheadRoberts2019; reading: passages §2.1 and §2.2, printed pp. 1291–1294, including Assumption 2.1, Theorem 2.2 and Proposition 2.3; author-hosted paper]. No sampling algorithm is being run here; the mechanical mapping and its scope follow from the written calculation above.

Required force, impulses and energy. The identities $\dot{a} = b^2 - a^2$, $\dot{b} = -2ab$ show that preserving the branch momentum between jumps requires

$$\begin{aligned} F_\sigma(q,t) &= m[\partial_t(a - \sigma b) + (a - \sigma b)^2]x = \frac{2\kappa^2 x}{mB^2}, \\ \Delta p_\sigma &= p_{-\sigma} - p_\sigma = 2m\sigma b x = \frac{2\kappa\sigma x}{B}. \end{aligned} \tag{148}$$

The force is independent of sign but not of preparation. It is generated by the prescribed time-dependent potential $V(q,t) = -\kappa^2 x^2/(mB(t)^2)$. For a fixed pre-jump sign, the impulse is a canonical momentum translation generated by $G_\sigma(q,t) = -\kappa\sigma x^2/B(t)$. These prescribed actuators do not themselves supply a Hamiltonian apparatus implementing the sign switch, its state-dependent clock or the required memory.

Under the stationary preparation, $Ex^2 = B$, $Ex^4 = 3B^2$ and the sign is independent of position. The mean continuous power and jump power are

$$\begin{aligned} E[F_\sigma \dot{Q}] &= 2mab^2 B, \\ E[\lambda_\sigma \Delta(p_\sigma^2/(2m))] &= mabE[x^2(\lambda_+ - \lambda_-)] = -2mab^2 B. \end{aligned} \tag{149}$$

Here $\Delta(p_\sigma^2/(2m)) = 2m\sigma abx^2$ and $E[x^2(z^2-1)] = 2B$. Their cancellation keeps the *ensemble mean* kinetic energy $\kappa^2/(2mA)$ fixed. Individual trajectories exchange energy with the prescribed actuator. No bounded reservoir, pathwise total-energy closure or apparatus record law has been established.

The minimal rate has the sharp expected total switch count

$$EN_T = 2\varphi(1)\arctan\frac{\kappa T}{mA}, \qquad EN_\infty = \pi\varphi(1) < 1. \tag{150}$$

Indeed $E|Z^2-1| = 4\varphi(1)$: integration by parts gives $E(Z^2-1)_+ = 2\varphi(1)$, and its zero mean makes the negative part equal. The expected rate is therefore $2b\varphi(1)$; integrate using (147). Both (150) and the pathwise bound concern the actual retained switch history, not a fresh noise allocation per observation.

Proposition 22 (microscopic force alone cannot supply this repair). Assume locally finite sign jumps, differentiable motion between jumps, ordinary velocity p/m, no other momentum changes between jumps, and a continuous force $F(t,q,p,\sigma)$ independent of the preparation. It cannot realize (145)'s branch law for all Gaussian means and widths.

Proof. Fix a common initial time and microscopic state $(q_0,p_0,+)$ with $p_0 \neq 0$. For any $A > 0$, choose $\mu_A = q_0 + p_0 A/\kappa$. This same state lies on the initial positive branch $p = -\kappa(q-\mu_A)/A$. Formula (148) requires

$$F(0,q_0,p_0,+) = -\frac{2\kappa p_0}{mA}. \tag{151}$$

Different widths give different forces. Each branch has a strictly positive position density. Continuity extends the required force from almost-everywhere agreement on that branch to the selected point, so the single microscopic force cannot satisfy both preparations. Explicit time dependence does not remove the common-time contradiction. □

This obstruction permits forces depending on retained width, mean, density or additional apparatus state. It closes only the proposal that the body pair and one sign already form the autonomous state. Unread preparations must retain their distinct controller labels and mean energy; replacing them by one marginal-score controller would again erase (119)'s excess. Also $Q_t - \mu$ never changes sign in this realization: matching a Gaussian marginal does not supply packet crossing or relative-phase reunion. The complete ideal pointer-readout escape (140) is unchanged.

7. The controller is the field pair

The realization needs four controller functions, $F_\pm(q,t)$ and $\lambda_\pm(q,t)$. Proposition 22 showed that they cannot be a preparation-independent law on (t,q,p,σ). The following identification says exactly what they are.

Proposition 9 (controller identification). Work in the class of smooth positive ρ with $\rho, \rho_q \to 0$ at infinity and known U. (a) At one time, the rate difference determines ρ: with $W = (m/\kappa)(\lambda_- - \lambda_+)$, two positive decaying solutions of $\rho_{qq} = W\rho$ have a constant Wronskian that tends to zero, hence are proportional, and normalization fixes the factor. (b) Over any open time interval, the rates determine $\rho(\cdot,t)$, hence ρ_t; with the flux hypothesis $\rho v \to 0$ at $-\infty$ and the displayed integral finite, continuity gives $\rho v = -\int_{-\infty}^{q} \rho_t\, dq'$, and finite kinetic energy fixes the constant in any case, since two currents differing by C would need $C^2 \int dq/\rho < \infty$. So they determine v, and S modulo a spatial constant whose time dependence (4) fixes. The forces are then fixed by (27) and add no data. (c) The forces alone also determine the field pair over an interval: $F_+ - F_- = -2\kappa v_{qq}$ gives v modulo affine functions; $\frac{1}{2}(F_+ + F_-) + U_q = (\kappa^2/m)\partial_q(\rho_{qq}/\rho)$ gives ρ_{qq}/ρ up to a constant c, and two positive decaying solutions of $\rho'' = (W_0 + c_i)\rho$ with $c_1 \neq c_2$ would have Wronskian increment $(c_2 - c_1)\int \rho_1\rho_2\, dq \neq 0$ between $-\infty$ and $+\infty$, contradicting decay; continuity then fixes the affine part of v. (d) At $\kappa = 0$ the minimal rates vanish, common flips remain allowed, and $F_\sigma = -U_q$ for every preparation: the zero branch admits a preparation-independent body law. For $\kappa > 0$ no such law exists for all Gaussian preparations under the assumptions of Proposition 22 of the recording note; that obstruction leaves open a fixed preparation, extra apparatus variables, or a different realization of the field law.

Proof. (a) If $\rho_1, \rho_2 > 0$ solve $\rho'' = W\rho$ then $w = \rho_1\rho_2' - \rho_1'\rho_2$ has $w' = 0$, and $w \to 0$ at infinity by decay, so $(\rho_2/\rho_1)' = w/\rho_1^2 = 0$. (b) Continuity $\rho_t = -\partial_q(\rho v)$ with the flux hypothesis, or the kinetic-energy argument. (c) The displayed combinations follow from (27); the Wronskian computation is $w' = \rho_1\rho_2'' - \rho_1''\rho_2 = (c_2 - c_1)\rho_1\rho_2$. (d) Insert $d = 0$ in (26)–(27). □

So either half of the controller is informationally the field pair, within this sheet architecture. An apparatus implementing Proposition 7 stores (ρ, S), or something from which (ρ, S) is computed, and updates it by (4). That is the pilot-wave structure: the field is a dynamical variable of the single system, carried alongside the particle. The known finite closed realization of the same law is the interacting-ensemble one, in which finitely many classical copies interact through a force built from the Fisher information of their empirical density and the field is recovered only as the copy number grows [@HallDeckertWiseman2014; reading: metadata via Crossref and arXiv abstract]. There the controller datum is the configuration of the other copies and κ is the interworld coupling constant. In both realizations κ is a coupling that nothing in the construction fixes.

Corollary 10 (one record leaves the bounded-degree family). Let $\mathcal{C}$ be the family of Gaussian-polynomial field pairs of bounded degree containing (18). One quadratic coordinate copy of (18) with record r produces, by Proposition 5, a pair (ρ_r, S) outside $\mathcal{C}$, and by Proposition 9 the controller functions on any interval after the copy determine that pair; so a controller confined to $\mathcal{C}$ cannot continue. This does not exclude finite-parameter controllers: the family of normalized pairs $(\rho_\epsilon(t)e^{-(r-q^2/\ell)^2/(2C)}, S_\epsilon(t))$ with parameters $(t, \epsilon, A, r, C, \ell)$ contains every one-copy posterior, and n identical quadratic copies at one time enter only through the sufficient statistics $(n, \sum_i r_i)$, the normalization absorbing $\sum_i r_i^2$. The controller must carry the record-dependent data of its family, not the literal history.

Proposition 11 (two unread widths, with a coordinate witness). Let a label $J \in \{1, 2\}$ with weights $\frac{1}{2}$ select preparations $N(0, A_j)$, $S = 0$, $A_1 \neq A_2$, and let J be unread. (a) *Branch realization.* Two controllers (145)–(148), one per A_j, give the mixture process; the sign stays fair given (Q, J), the mean kinetic energy is $\frac{\kappa^2}{4m}(A_1^{-1} + A_2^{-1})$ for all time, and

$$EQ_t^2 = \frac{A_1 + A_2}{2} + \frac{\kappa^2 t^2}{2m^2}\Big(\frac{1}{A_1} + \frac{1}{A_2}\Big). \tag{31}$$

(b) *Single marginal-score controller.* Starting one controller from $\bar{\rho} = \frac{1}{2}(\rho_1 + \rho_2)$ and $\bar{S} = 0$ gives kinetic energy $\frac{\kappa^2}{2m} I(\bar{\rho})$ and, by the free virial identity $\frac{d^2}{dt^2} EQ^2 = 4\mathcal{H}_{\mathrm{kin}}/m$, $EQ_t^2 = \frac{1}{2}(A_1 + A_2) + \kappa^2 I(\bar{\rho}) t^2/m^2$. The two differ by

$$\frac{\kappa^2 t^2}{m^2}\Delta, \qquad \Delta = \frac{1}{2}\Big(\frac{1}{A_1} + \frac{1}{A_2}\Big) - I(\bar{\rho}) = \Big(\frac{1}{A_1} - \frac{1}{A_2}\Big)^2 \int \frac{q^2 \rho_1 \rho_2}{4\bar{\rho}}\, dq > 0, \tag{32}$$

which is (119) for this mixture. The single controller loses energy $\kappa^2\Delta/(2m)$ and underpredicts the variance by $\kappa^2 \Delta t^2/m^2$ at every $t > 0$: a difference visible in coordinate records alone.

Proof. (a) Each branch is Proposition 21 with its own A_j; the energies and variances are (143)'s, averaged. (b) The virial identity is the time derivative of (19)'s structure for any free solution of (4): $\frac{d}{dt}\int \rho q^2 = 2\int \rho q v$ and $\frac{d^2}{dt^2}\int \rho q^2 = 2\int \rho(v^2 + d^2)\, dq$ by (25) and two integrations by parts, which is $4\mathcal{H}_{\mathrm{kin}}/m$. For Δ, the conditional label law is $\pi_j(q) = \rho_j/(2\bar{\rho})$ and the branch scores are $-q/A_j$, so $\mathrm{Var}(\sigma_J(q) \mid q) = q^2 \pi_1 \pi_2 (A_1^{-1} - A_2^{-1})^2$; insert in (119). □

Within this model, the rule against replacing a discarded label by a fresh marginal score is therefore a theorem with a coordinate witness, under one qualification: the single-controller comparison uses the algebraic evolution (15) of $\sqrt{\bar{\rho}}$, whose positivity and nonexplosion as a sheet path law through possible nodes are not proved here, while the virial comparison needs only (15). It rules out this marginal-score replacement, not compressed descriptions of the branches. Closure under unread records holds for branch controllers, which carry the record-dependent data of Corollary 10.

Proposition 12 (closure within the theory, at every κ). Coordinate shears $f(q)\pi$ preserve the class (2)/(136) with its fair sign (Proposition 20 of the recording note), and the sheet dynamics of Proposition 7, applied to body and pointers alike with their own potentials, preserves it by construction wherever $\rho > 0$. (For several configuration coordinates with one

shared sign and masses m_b, Propositions 6–7 hold with $\lambda_- - \lambda_+ = \kappa \sum_b \rho_{bb}/(m_b\rho)$ and $F_{\sigma,a} = -\partial_a[U - \kappa^2 \sum_b \rho_{bb}/(m_b\rho) + \sigma\kappa \sum_b S_{bb}/m_b]$: the cross terms cancel by $m_b\partial_a v_b = m_a\partial_b v_a$ and its analogue for d, which hold because $v_b = S_b/m_b$ and $d_b = \kappa L_b/m_b$.) Hence every finite composition of coordinate copies, sheet evolution and conditioning on coordinate records stays in the class, provided the positive conservative path evolutions it requires exist, which Proposition 8 establishes for its orbit and not for all posteriors, and provided the protocol evolves each conditioned branch with the controller of its own posterior field pair: conditioning an already running process on past records does not by itself replace its drift and rates, since (22) changes the score by $(r - f)f'/C$ while a preassigned controller is unchanged. Under that branch-dependent protocol the floor (138) holds for every complete coordinate record. The escape (140) reads a canonical momentum directly; within the theory a momentum is accessed only through coordinate copies after sheet evolution, which keeps the floor, and the exclusion of direct conjugate access is a stipulated measurement rule. At $\kappa = 0$ the minimal-switch single-sheet construction closes before caustics, where (138) reads $\det \geq 0$. Closure under composition and complete coordinate records, where it holds, holds for every $\kappa \geq 0$ and selects no scale. Neither branch is unconditionally closed: the $\kappa = 0$ single-sheet class leaves itself at caustics of the classical flow, and the $\kappa > 0$ realization needs $\rho > 0$ and is not continued through nodes here.

Consequence for STATE

The exact sheet restriction, positive forward orbit, controller identification, copy-generated tails and unread-width witness retain their respective hypotheses. A finite bounded-degree Hermite family fails after one nonlinear record; finite-parameter controllers with sufficient statistics remain possible. Neither literal full-history storage nor impossibility of every realization has been proved. Branch-dependent controller updates and positive conservative path evolutions are required for the stated copy/evolution closure. Nodes and classical caustics remain separate boundaries. The zero branch is admitted where its single-sheet flow exists, so this construction supplies no independent positivity, universality or identification with physical $\hbar$.